

Memoria Práctica 1

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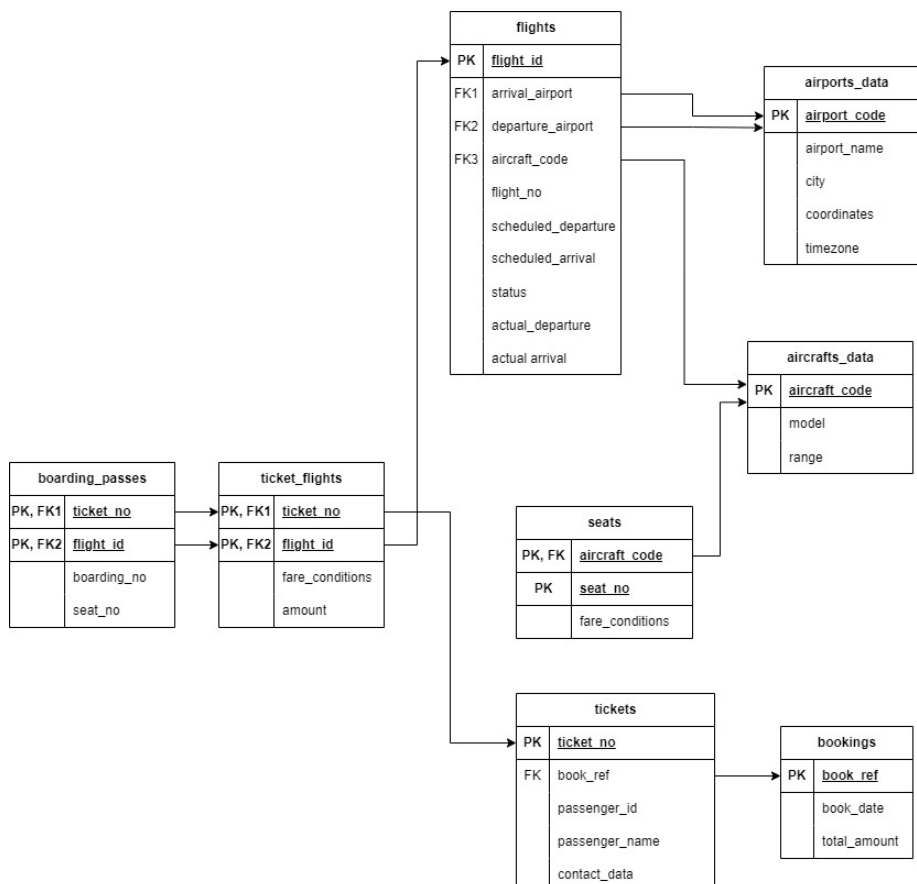
Familiarizarse con la Base de Datos

Para examinar las claves de las tablas de la base de datos creada, usamos el cliente gráfico DBeaver.

A continuación, se muestra el esquema de la base de datos:

- aircrafts_data(**aircraft_code**,model,range)
- airports_data(**airport_code**,airport_name,city,coordinates,timezone)
- boarding_passes(**ticket_no**, **flight_id**,ticket_no->ticket_flights.ticket_no,flight_id->ticket_flights.flight_id,boarding_no,seat_no)`
- bookings(**book_ref**, book_date, total_amount)
- flights(**flight_id**,aircraft_code->aircrafts_data.aircraft_code,arrival_airport->airports_data.airport_code, departure_airport->airports_data.airport_code,flight_no,scheduled_departure, scheduled_arrival, status, actual_departure, actual_arrival)
- seats(**aircraft_code**, **seat_no**,aircraft_code->aircrafts_data.aircraft_code, fare_conditions)
- ticket_flights(**ticket_no**, **flight_id**,flight_id->flights.flight_id, ticket_no->tickets.ticket_number,fare_conditions, amount)
- tickets(**ticket_no**,book_ref->bookings.bookref, passenger_id, passenger_name, contact_data`

Diagrama del modelo relacional de la base de datos implementada



Consultas

Consulta 1 (lo que llevamos)

No fuimos capaces de hacerla

```
SELECT bookings.book_ref, tickets.ticket_no, flights.flight_id,
       flights.departure_airport, flights.arrival_airport
FROM bookings
JOIN tickets ON bookings.book_ref = tickets.book_ref
JOIN ticket_flights ON tickets.ticket_no = ticket_flights.ticket_no
JOIN flights ON ticket_flights.flight_id = flights.flight_id;
```

Consulta 2

Para hacer esta consulta hicimos una tabla con el numero de reserva, el monto total de la reserva y una tercera columna que se corresponde con la suma del monto de todos los billetes de una reserva. Hicimos esto para todas las reservas.

```
SELECT bookings.book_ref,
       bookings.total_amount,
       Sum(ticket_flights.amount) AS total
FROM bookings
LEFT JOIN tickets
      ON bookings.book_ref = tickets.book_ref
JOIN ticket_flights
      ON tickets.ticket_no = ticket_flights.ticket_no
GROUP BY bookings.book_ref
ORDER BY bookings.book_ref ASC;
```

Consulta 3

Para esta consulta hicimos una tabla con el codigo de todos los aeropuertos y una segunda columna (Pasajeros) que cuenta el numero de flight_id en boarding_passes por cada aeropuerto de destino

```
SELECT a.airport_code,
       Count(bp.flight_id) AS Pasajeros
FROM airports_data a
JOIN flights f
      ON f.arrival_airport = a.airport_code
JOIN ticket_flights tf
      ON tf.flight_id = f.flight_id
JOIN boarding_passes bp
      ON tf.ticket_no = bp.ticket_no
GROUP BY a.airport_code
ORDER BY pasajeros ASC;
```

Consulta 4

Para esta consulta calculamos el numero de asientos totales en cada avión asociado a cada flight_id, después calculamos el numero de asientos reservados contando los numeros de ticket asociados a un flight_id, lo restamos y mostramos el resultado mas alto en una tabla junto con el flight_id

```
WITH asientos_totales
  AS (SELECT f.flight_id,
            Count(s.seat_no) AS asientos_totales
      FROM flights f
      JOIN aircrafts_data ad
        ON ad.aircraft_code = f.aircraft_code
      JOIN seats s
        ON s.aircraft_code = ad.aircraft_code
      GROUP BY f.flight_id),
asientos_reservados
  AS (SELECT f.flight_id,
            Count(tf.ticket_no) AS asientos_reservados
      FROM ticket_flights tf
      FULL JOIN flights f
        ON f.flight_id = tf.flight_id
      GROUP BY f.flight_id),
asientos_vacios
  AS (SELECT at.flight_id,
            ( at.asientos_totales - ar.asientos_reservados ) AS vacios
      FROM asientos_totales at
      FULL JOIN asientos_reservados ar
        ON ar.flight_id = at.flight_id)
SELECT av.flight_id,
       av.vacios
FROM asientos_vacios av
WHERE av.vacios = (SELECT Max(av.vacios)
                  FROM asientos_vacios av);
```

Consulta 5

Para esta consulta creamos dos subtablas: la primera reúne las reservas con tarjetas de embarque, y la segunda subtabla reúne la totalidad de las reservas. Juntamos las dos subtablas tal que las tuplas de la primera subtabla no tengan ninguna tarjeta de embarque

```
WITH reservas_totales
  AS (SELECT tf.ticket_no,
            tf.flight_id,
            t.book_ref
      FROM ticket_flights tf,
            tickets t
      WHERE tf.ticket_no = t.ticket_no),
  reservas_sin_tarjetas_de_embarque
  AS (SELECT tf.flight_id,
            t.book_ref,
            tf.ticket_no
      FROM boarding_passes bp,
            tickets t,
            ticket_flights tf
      WHERE bp.ticket_no = t.ticket_no
            AND t.ticket_no = tf.ticket_no
            AND bp.flight_id = tf.flight_id)
SELECT DISTINCT rt.book_ref,
               rt.flight_id
FROM   reservas_totales rt
      LEFT JOIN reservas_sin_tarjetas_de_embarque rstde
            ON rt.book_ref = rstde.book_ref
               AND rt.flight_id = rstde.flight_id
               AND rt.ticket_no = rstde.ticket_no
WHERE  rstde.book_ref IS NULL
       OR rstde.ticket_no IS NULL
ORDER BY rt.book_ref,
         rt.flight_id;
```

Consulta 6

Para esta consulta hicimos una tabla con el numero de vuelo y una columna con el promedio de retraso de todos los vuelos asociados a un numero de vuelo y mostramos el resultado mas alto

```
WITH retraso_medio
  AS (SELECT f.flight_no,
            Avg(f.actual_arrival - f.scheduled_arrival) AS
retraso_medio
      FROM   flights f
      GROUP BY f.flight_no
      ORDER BY retraso_medio DESC)
SELECT rm.flight_no,
       rm.retraso_medio
FROM   retraso_medio rm
WHERE  rm.retraso_medio = (SELECT Max(rm.retraso_medio)
                        FROM   retraso_medio rm);
```